

MDS5202 Assignment 3 Report

Course Assistant

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Conceptual

5. LDA vs. QDA

- (a) **Training:** QDA is more flexible, so it usually fits the training set at least as well as LDA. **Test:** when the Bayes boundary is linear, LDA tends to generalize better because of lower variance.
- (b) **Training:** QDA typically achieves lower training error. **Test:** if the true boundary is nonlinear and the sample size is large, QDA is often superior; with limited data, LDA can be more stable.
- (c) As n grows, QDA estimates become more reliable, so its test performance relative to LDA generally improves or stays similar.
- (d) **False.** Even when the optimal boundary is linear, QDA may overfit due to higher variance and can have worse test error than LDA.

6. Logistic regression probability

- (a) Linear score: $\eta = -6 + 0.05 \times 40 + 1 \times 3.5 = -0.5$. Probability: $p = \frac{1}{1+e^{-\eta}} = \frac{1}{1+e^{0.5}} \approx 0.3775$.
- (b) For $p = 0.5$, we have $\eta = 0$: $-6 + 0.05x + 3.5 = 0 \Rightarrow 0.05x = 2.5 \Rightarrow x = 50$ hours.

Applied

13(a) Numerical and graphical summaries

Counts: Up = 605, Down = 484. Correlations between lagged returns and Today are small (e.g., $\text{corr}(\text{Lag2}, \text{Today}) \approx 0.059$).

Figures: see Figures 1–3.

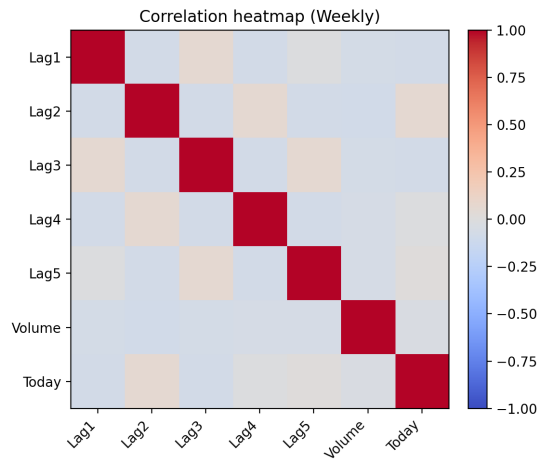


Figure 1: Correlation heatmap for Lag/Volume/Today

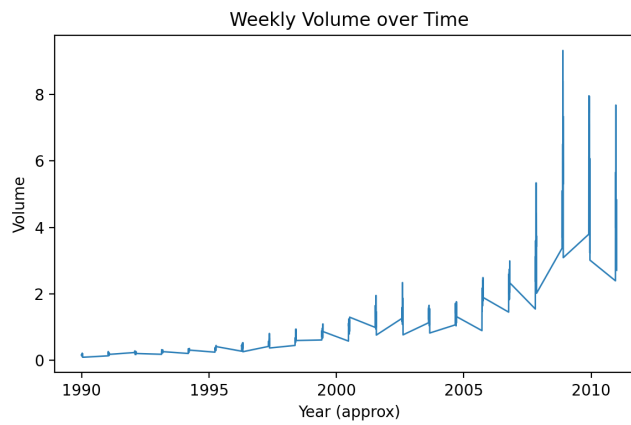


Figure 2: Weekly trading volume over time

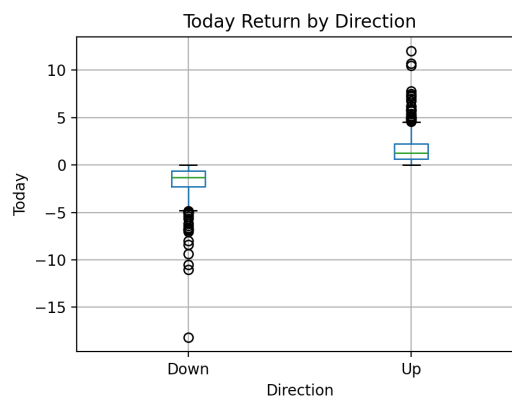


Figure 3: Today by market direction

Takeaway: lagged returns show weak linear association with Today; Volume trends upward; Up weeks are slightly more frequent.

13(b) Full-data logistic regression significance

Only **Lag2** is significant at the 5% level ($p \approx 0.03$). Other predictors are not statistically significant, indicating limited linear predictive signal.

13(c) Full-data confusion matrix

	Pred Down	Pred Up
Actual Down	54	430
Actual Up	48	557

Table 1: Full-data logistic regression confusion matrix (threshold 0.5), accuracy ≈ 0.561

Interpretation: the model favors Up predictions, leading to many Down weeks being classified as Up (FP=430). Overall accuracy is about 56%.

13(d)–(h) Train 1990–2008, test 2009–2010 (Lag2 only)

Method	TN	FP	FN	TP
Logistic	9	34	5	56
LDA	9	34	5	56
QDA	0	43	0	61
KNN (k=1)	21	22	30	31
Naive Bayes	0	43	0	61

Table 2: Test-set confusion matrices (2009–2010); accuracies are 0.625, 0.625, 0.5865, 0.500, 0.5865

13(i) Best method (based on 13(d)–(h))

Logistic regression and LDA tie for the best test accuracy at 0.625.

13(j) Exploring predictor combinations and transformations

Among tried combinations, a KNN model with $k = 5$ using Lag2 and Lag3² performs best:

KNN (k=5) with predictors: Lag2 and Lag3², test accuracy ≈ 0.6827 , confusion matrix:

	Pred Down	Pred Up
Actual Down	25	18
Actual Up	15	46

Table 3: Best-performing combination (test set)

15. Functions (Python version)

(a)–(d) Power functions and values

```

# (a) Power(): 2^3

def Power():
    print(2**3)

# (b) Power2(x, a): print x^a

def Power2(x, a):
    print(x**a)

# (d) Power3(x, a): return x^a

def Power3(x, a):
    result = x**a
    return result

# (c) Compute
print(Power3(10, 3))    # 1000
print(Power3(8, 17))    # 2251799813685248
print(Power3(13, 13))   # 302875106592253

```

Values: $10^3 = 1000$, $8^{17} = 2251799813685248$, $13^{13} = 302875106592253$.

(e) Plot of $f(x) = x^2$

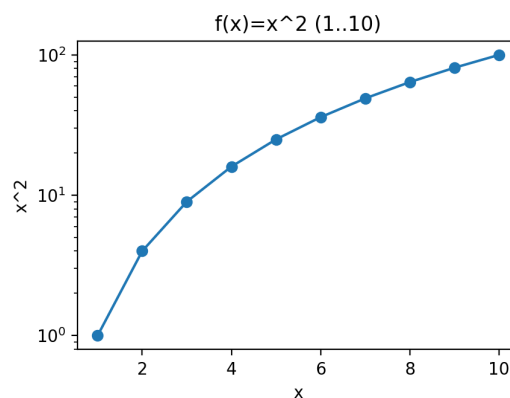


Figure 4: $f(x) = x^2$ ($x = 1..10$, y-axis on log scale)

(f) PlotPower() function

```

import numpy as np
import matplotlib.pyplot as plt

def PlotPower(x, a):
    x = np.asarray(x)
    y = x**a
    plt.plot(x, y, marker='o')
    plt.xlabel('x')
    plt.ylabel(f'x^{a}')
    plt.title(f'PlotPower(x, a={a})')
    plt.show()

# Example

```

```
PlotPower(range(1, 11), 3)
```

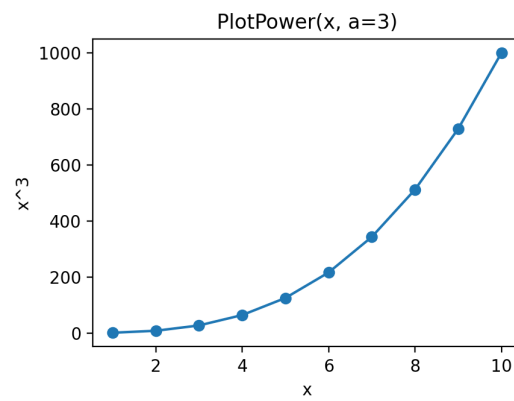


Figure 5: Example: PlotPower(1..10, 3)